

Spreading 33 Points in the Plane

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October 8, 2026

Abstract

We place 33 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 29.80600+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 34-point configuration gives 30.90471; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 33 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 29.80600\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 33$ is not on the page. Deleting a point from our 34-point configuration gives 30.90471; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n33.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
7451502112639781960841389
denominator:
249999999999626064249796
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

33 points, x and y per line. File: friedman-maxmin2d/coords/n33.txt.

```
-6.689370643687 2.245958332565 -6.260470196984 3.149310095068
-3.987842814420 -1.154745821527 -1.405825405985 2.424838138453
-3.187564973900 1.770387635146 -2.715063571990 2.651717553617
-4.017305934596 -0.155179953487 -5.855641304015 -0.522343659209
-2.153558914868 -0.447812111833 -5.004723429625 0.002955085358
-5.358279064378 2.717973613596 -4.975260309449 -0.996610782681
-3.592389703256 0.855993333708 -3.945942520073 2.422203174591
-3.621843497499 4.292460154987 -2.188060174434 1.801854377003
-4.510181931501 3.247814394591 -4.590749547705 0.913243844538
-6.432516202870 0.294488851818 -1.307679179867 1.327587253531
-4.794039652950 1.892362393596 -2.446727542273 0.835887921677
-1.932828803541 3.274701315067 -5.790460062126 1.807826199117
-4.620554607071 4.241704666446 -5.587169956882 0.828707650059
-5.535469796616 3.838058539325 -3.004476789258 -0.973110856400
-3.473441118162 3.303533093062 -1.566346547706 0.361620798205
-6.736763308452 1.247081996211 -2.691206349714 3.926516854512
-3.033939909434 0.026455011639
```